

Insert new Annex 179A through Annex 179D corresponding to 179 as follows:

Annex 179A

(informative)

Transmitter, receiver, and channel parameters associated with test points TP0d and TP5d for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8

179A.1 Overview

This annex provides information on transmitter, receiver, and channel parameters associated with test points TP0d and TP5d, which might not be testable in an implemented system. TP0d and TP5d are defined in 179.8.1 and illustrated in Figure 179–2. This annex also provides information on the channel (TP0d-to-TP5d).

179A.2 Transmitter characteristics at TP0d

The recommended transmitter characteristics at TP0d as measured at TP0v are described in 178.9.2. TP0d is depicted in Figure 179–2.

179A.3 Receiver characteristics at TP5d

The recommended receiver characteristics at TP5d as measured at TP5v are described in 178.9.3. TP5d is depicted in the link block diagram of Figure 179–2.

179A.4 Host channel insertion loss

The host channel, consisting of controlled impedance PCB, device package, and mated host connector, is illustrated in Figure 179A–1. The host channel does not include the TP2 or TP3 test fixture PCB.

For each host class, the recommended maximum differential insertion loss at 53.125 GHz for a host channel is given in Table 179A–1. The recommended minimum differential insertion loss at 53.125 GHz for a host channel is 3.45 dB.

The recommended maximum differential insertion loss at 53.125 GHz for (TP0d-to-TP2) or (TP3-to-TP5d) for each of the host channels is given in Table 179A–1, and is illustrated in Figure 179A–1 for a symmetric host nominal channel.

The recommended maximum differential insertion loss (TP0d-to-TP2) or (TP3-to-TP5d) is consistent with the host channels and the reference TP2 or TP3 test fixture specified in 179B.2.1.

Table 179A-1—Recommended differential insertion loss limits at 53.125 GHz

Host class	Host channels	TP0d to TP2 or TP3 to TP5d
	Max (dB)	Max (dB)
Host-Low (HL)	7.95	11.75
Host-Nominal (HN)	12.95	16.75
Host-High (HH)	17.95	21.75

The normative requirements for the host classes listed in Table 179A-1 are given in 179.9.

179A.5 Channel (TP0d-TP5d) insertion loss

This subclause provides information on the channel (TP0d-TP5d) insertion losses for the maximum-loss and minimum-loss cabling topologies.

The differential insertion loss of the channel (TP0d-TP5d) associated with the maximum insertion loss cable assembly and maximum insertion loss of (TP0d-TP2) and (TP3-TP5d) at 53.125 GHz is determined using Equation (179A-1). An example of the channel loss allocation for the HN-to-HN link configuration is illustrated in Figure 179A-2. The differential insertion loss of the channel (TP0d-TP5d) associated with the minimum insertion loss cable assembly and minimum insertion loss of (TP0d-TP2) and (TP3-TP5d) at 53.125 GHz is determined by Equation (179A-2) and illustrated in Figure 179A-3. Loss budget parameters at 53.125 GHz for all link configurations are provided in Table 179A-2 and Table 179A-3.

$$ILdd_{Ch,Max} = ILdd_{CA,Max} + ILdd_{TP0d-TP2,Max} + ILdd_{TP3-TP5d,Max} - 2 \times ILdd_{MTFref} \quad (179A-1)$$

$$ILdd_{Ch,Min} = ILdd_{CA,Min} + ILdd_{TP0d-TP2,Min} + ILdd_{TP3-TP5d,Min} - 2 \times ILdd_{MTFref} \quad (179A-2)$$

where

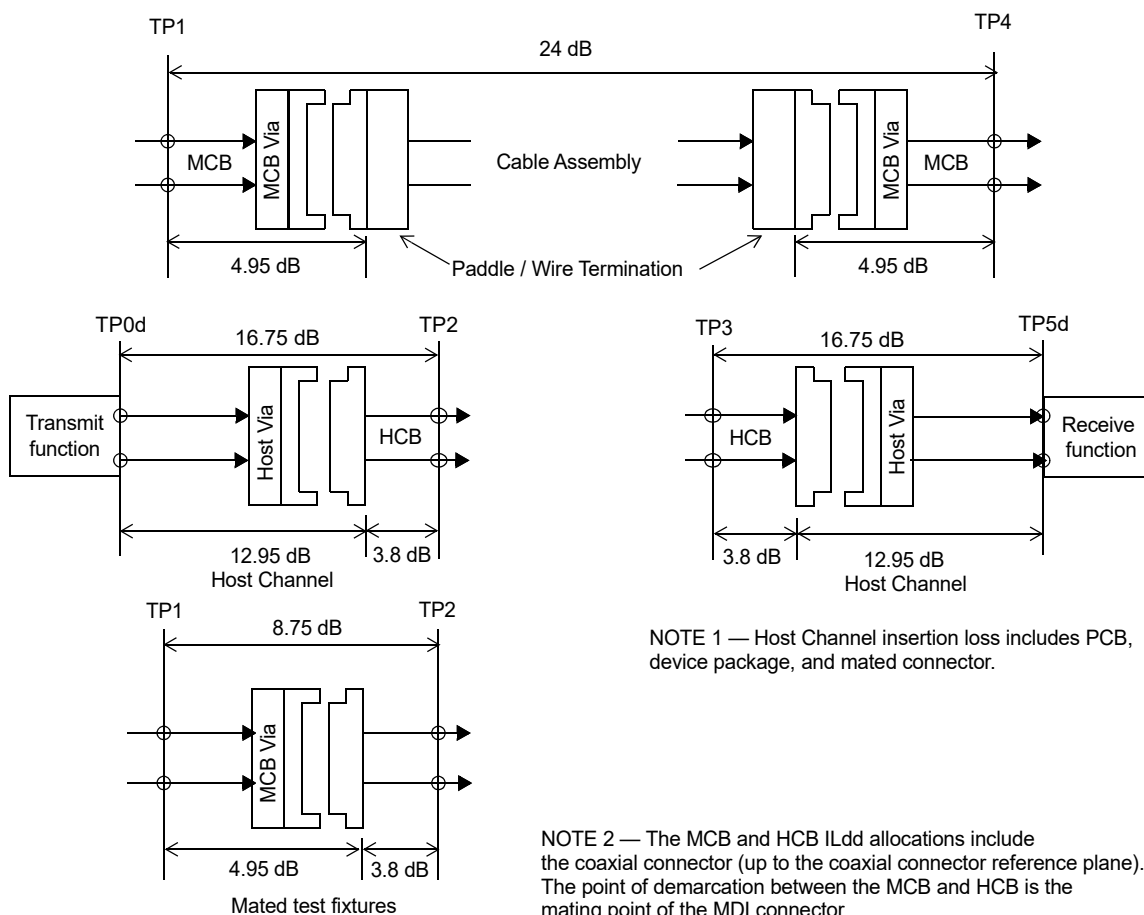
$ILdd_{Ch,Max}$	is the maximum channel insertion loss between TP0d and TP5d in dB
$ILdd_{Ch,Min}$	is the minimum channel insertion loss between TP0d and TP5d in dB
$ILdd_{CA,Max}$	is the maximum cable assembly insertion loss (TP1 to TP4) in dB, 179.11.2
$ILdd_{CA,Min}$	is the minimum cable assembly insertion loss (TP1 to TP4) in dB, 179.11.2
$ILdd_{TP0d-TP2,Max}$	is the maximum insertion loss from TP0d to TP2 in dB (Table 179A-2)
$ILdd_{TP3-TP5d,Max}$	is the maximum insertion loss from TP3 to TP5d in dB (Table 179A-2)
$ILdd_{TP0d-TP2,Min}$	is the minimum insertion loss from TP0d to TP2 in dB (Table 179A-3)
$ILdd_{TP3-TP5d,Min}$	is the minimum insertion loss from TP3 to TP5d in dB (Table 179A-3)
$ILdd_{MTFref}$	is the reference insertion loss of the mated test fixtures in dB, using Equation (179B-5)

Table 179A–2—Maximum Insertion loss budget values at 53.125 GHz

Link Configuration	$ILdd_{CA,max}$ (dB)	$ILdd_{TP0d-TP2,max}$ (dB)	$ILdd_{TP3-TP5d,max}$ (dB)	$ILdd_{MTFref}$ (dB)	$ILdd_{Ch,max}$ (dB)
HH to HN	19 (CA-A)	21.75	16.75	8.75	40
HH to HL	24 (CA-B)	21.75	11.75	8.75	40
HN to HN	24 (CA-B)	16.75	16.75	8.75	40
HN to HL	29 (CA-C)	16.75	11.75	8.75	40
HL to HL	34 (CA-D)	11.75	11.75	8.75	40

Table 179A–3—Minimum Insertion loss budget values at 53.125 GHz

Link Configuration	$ILdd_{CA,min}$ (dB)	$ILdd_{TP0d-TP2,min}$ (dB)	$ILdd_{TP3-TP5d,min}$ (dB)	$ILdd_{MTFref}$ (dB)	$ILdd_{Ch,min}$ (dB)
Host-Min to Host-Min	16	7.25	7.25	8.75	13



MCB = Module Compliance Board, 179B.3
HCB = Host Compliance Board, 179B.2
MCB Via = transition via to MDI connector on an MCB
Host Via = transition via to MDI connector on a Host Channel
Paddle/Wire Termination = transition structure(s) in a Cable Assembly not present on an HCB

Figure 179A-1—Host-Nominal to Host-Nominal, Cable assembly, and test fixture insertion loss at 53.125 GHz

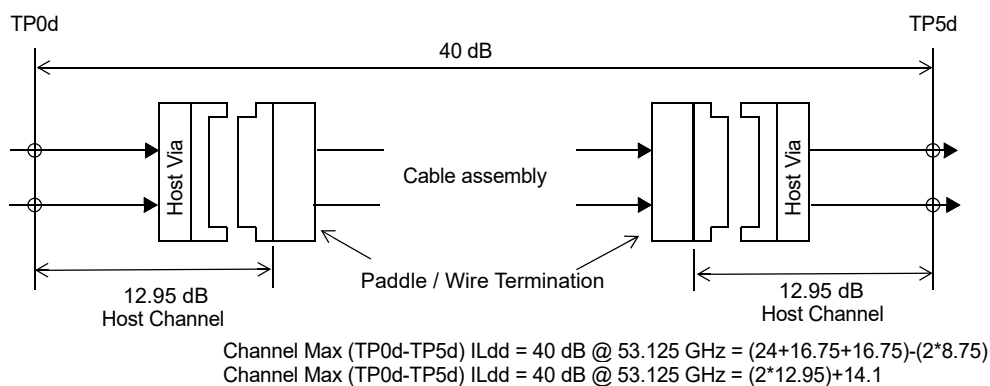


Figure 179A-2—HN-to-HN Channel Max (TP0d-TP5d) at 53.125 GHz

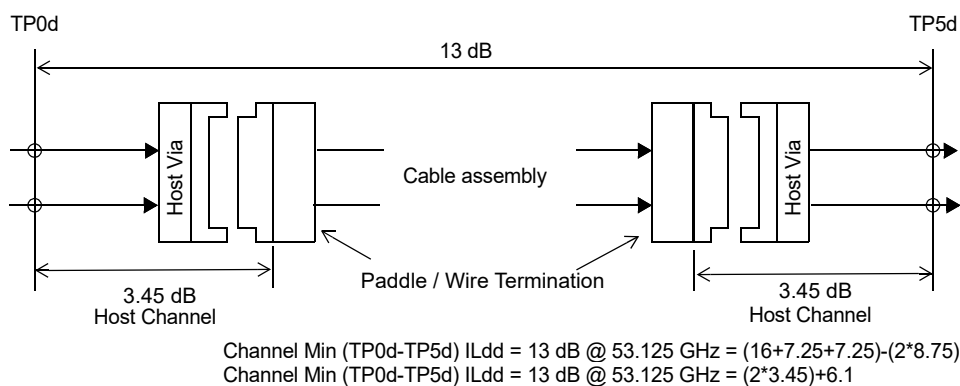


Figure 179A-3— Channel Min (TP0d-TP5d) at 53.125 GHz

Table 179A-4—Channel (TP0d-TP5d) configuration matrix for hosts and cable assemblies

Host class	Cable assembly classes compatible with HL partner	Cable assembly classes compatible with HN partner	Cable assembly classes compatible with HH partner
HL	CA-A,B,C,D	CA-A,B,C	CA-A,B
HN	CA-A,B,C	CA-A,B	CA-A
HH	CA-A,B	CA-A	not supported

179A.6 Channel (TP0d-TP5d) effective return loss (ERL)

The 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4 and 1.6TBASE-CR8 channels are recommended to meet the ERL specified in 178.10.3.

179A.7 Channel (TP0d-TP5d) Operating Margin (COM)

The Channel Operating Margin (COM) for the channel between TP0d and TP5d is computed using the using the maximum likelihood sequence detection (MLSD) defined in 178A.1.10 and the parameters in Table 178–15, and is recommended to be greater than or equal to 3 dB. Device package models are included in the TP0d-to-TP5d channel; COM (TP0d-to-TP5d) is computed with the concatenation of device termination models to the TP0d-to-TP5d channel.